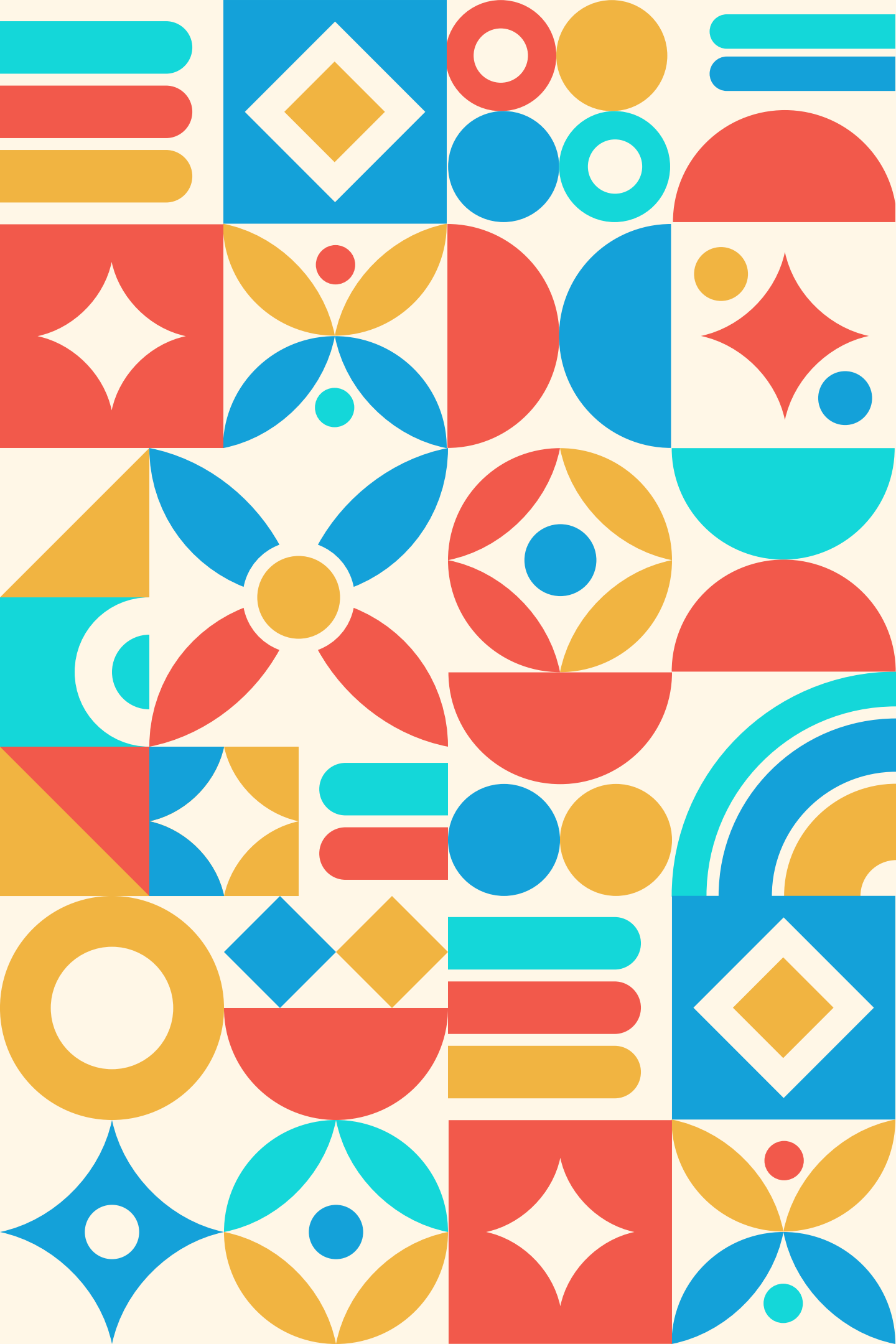




Mosaïque avec critères avancés

Master Imagine



**UNIVERSITÉ DE
MONTPELLIER**

Auteurs



ORAVEC
Tommy-Verdi

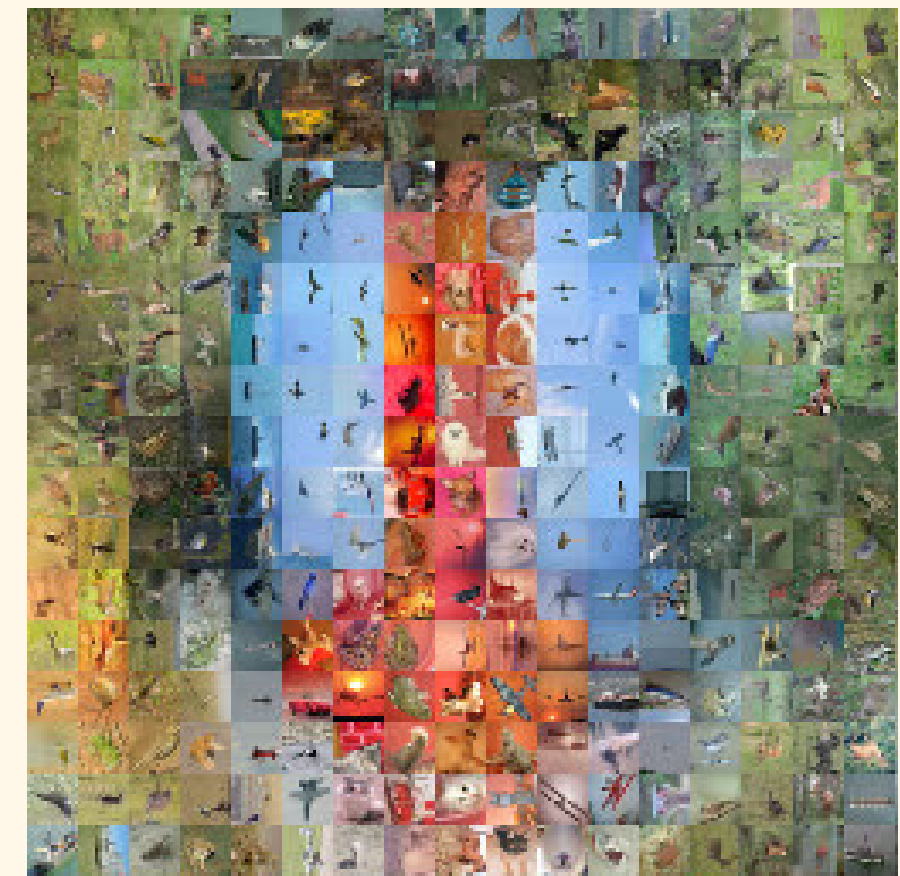
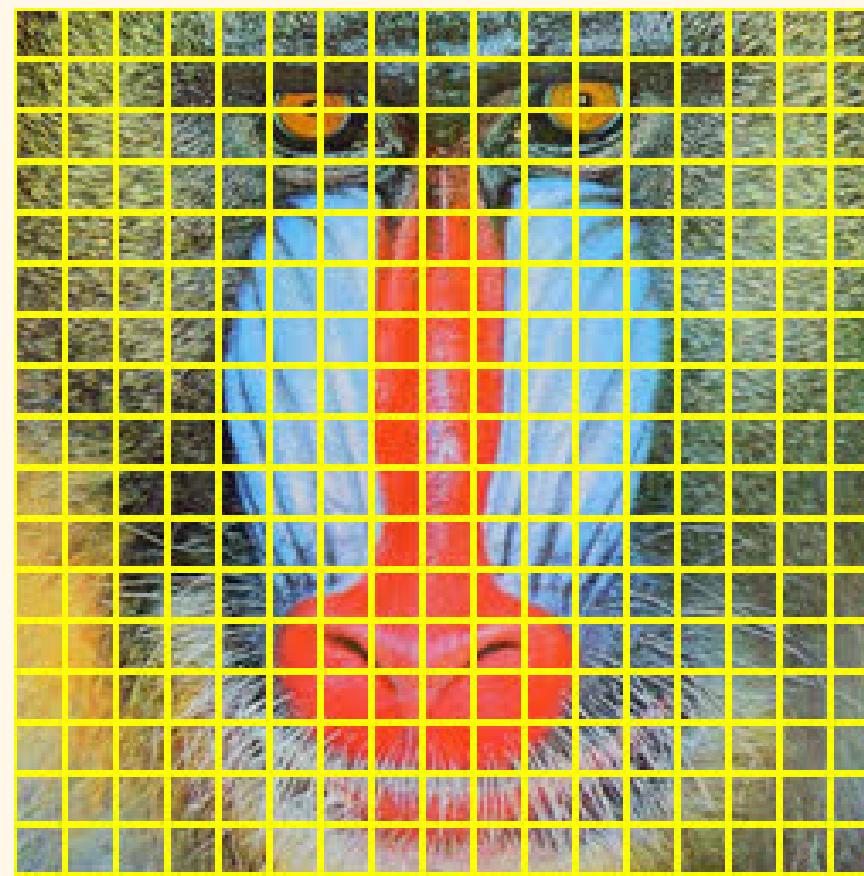
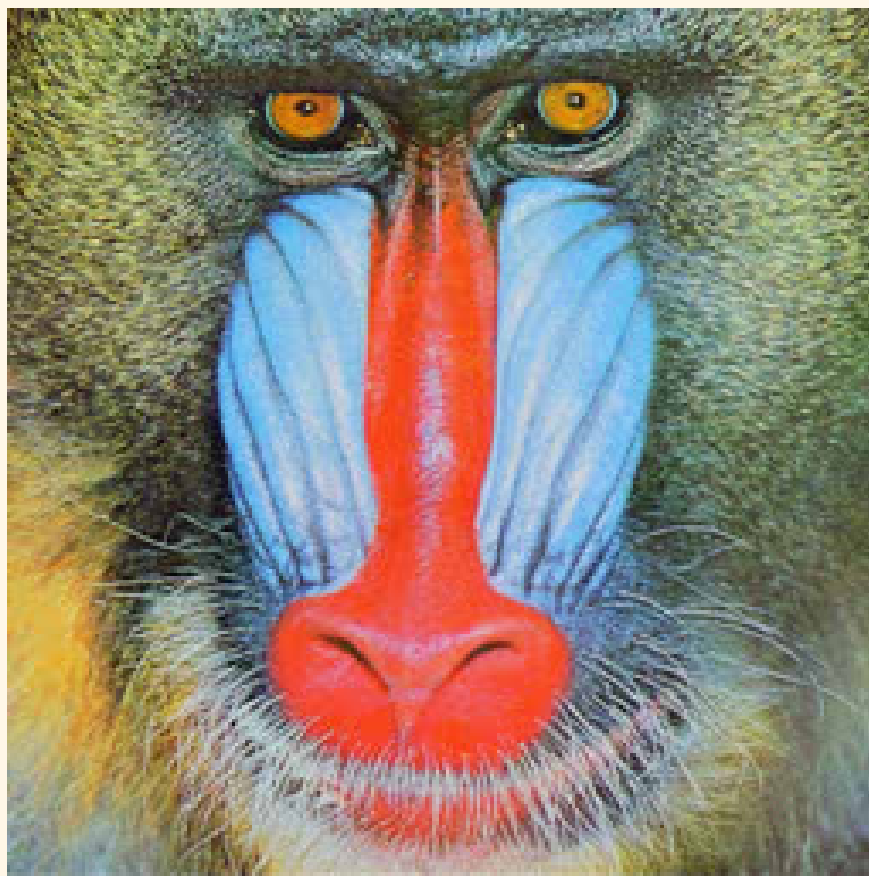


PAULO
Lucas



ONIC
Victor

La photomosaïque



**Classique
Adaptive tiling
Super-pixels**

Plan

1. Etat de l'art

- False photomosaic & Freeform mosaic
- Jigsaw mosaïque & Approximate Nearest Neighbor
- Autres approches : IA & deep learning

2. Méthodes implémentées

- Tuiles adaptatives (adaptive tiling, seuil de variance : 500)
- Super-pixels (SNIC)

3. Post-traitement

- 2nd pass et rotation des imageries (première passe → seconde passe)

4. Résultats

- Classique vs Classique + 2nd pass vs Adaptive tiling
- Super-pixel vs Super-pixel + 2nd pass

5. Analyses

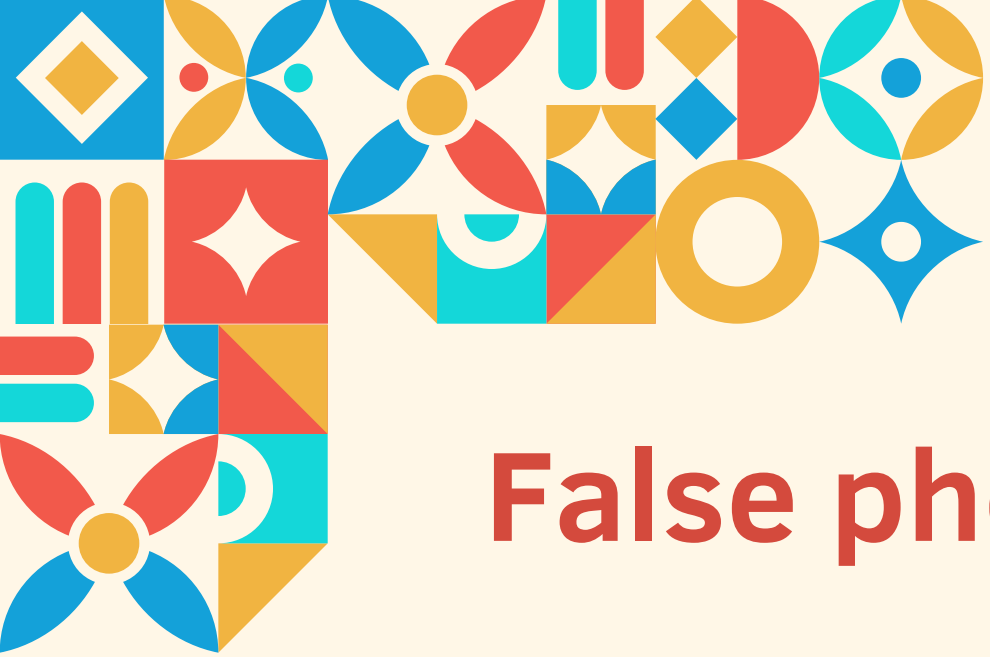
6. Conclusion

- Adaptive tiling : résultats insuffisants
- Mosaïque classique : stable
- Super-pixel : approche originale aux résultats variés

7. Perspective

- Amélioration de l'adaptive tiling, optimisation du temps de calcul, extension des angles de rotation, interface, espace couleurs, comparaisons 2nd pass

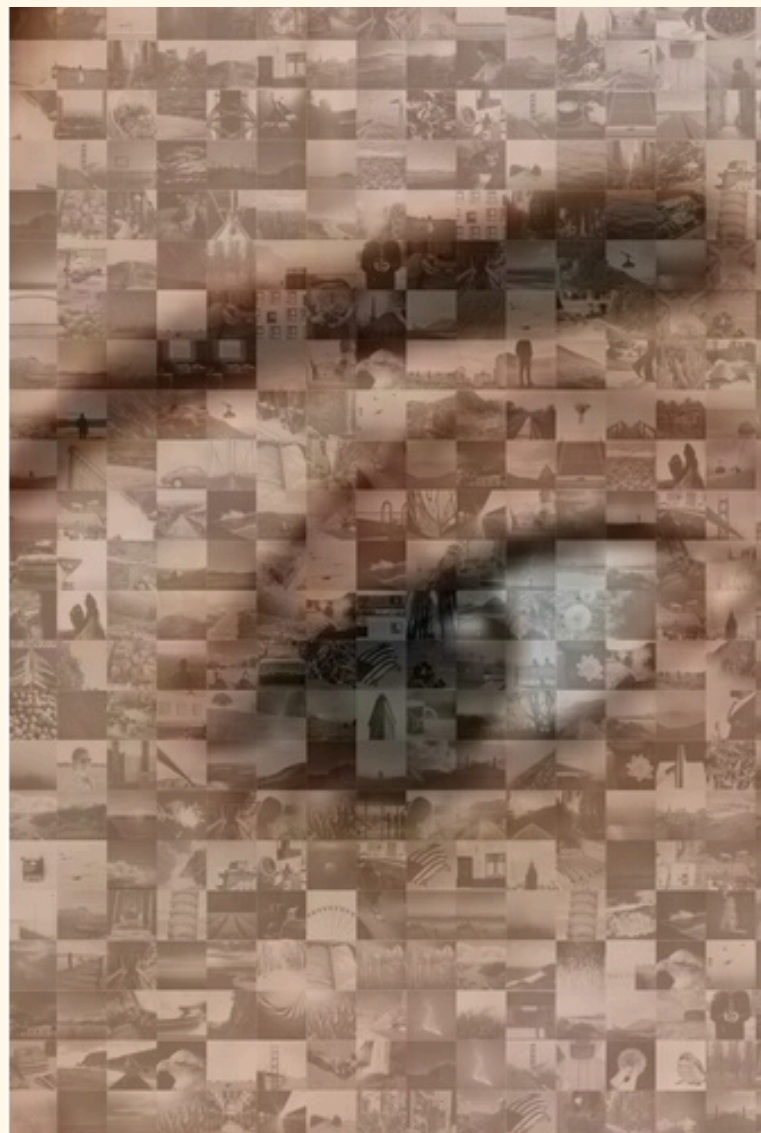
8. Sources



État de l'art



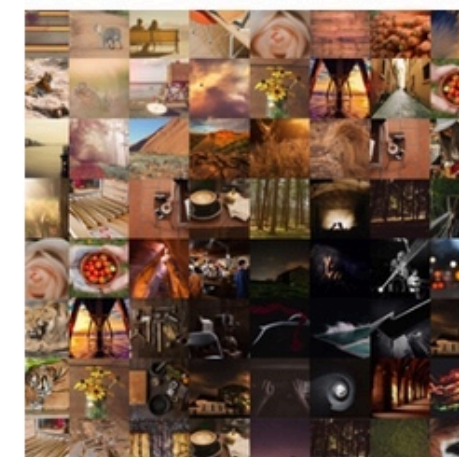
False photomosaic



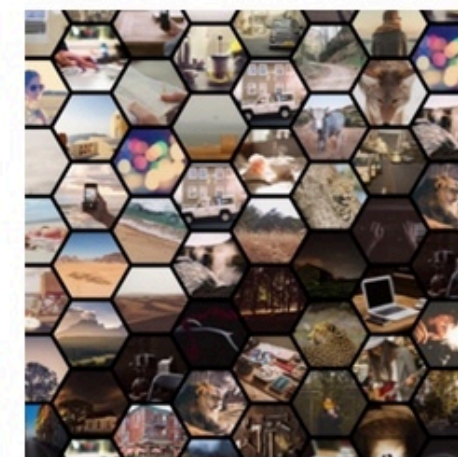
[1]



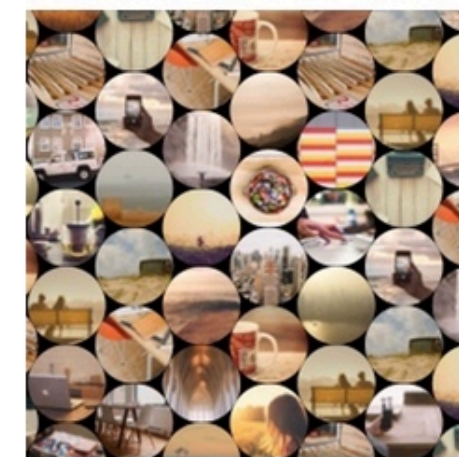
Freeform mosaic



Rectangular



Hexagonal



Circular



Diamond



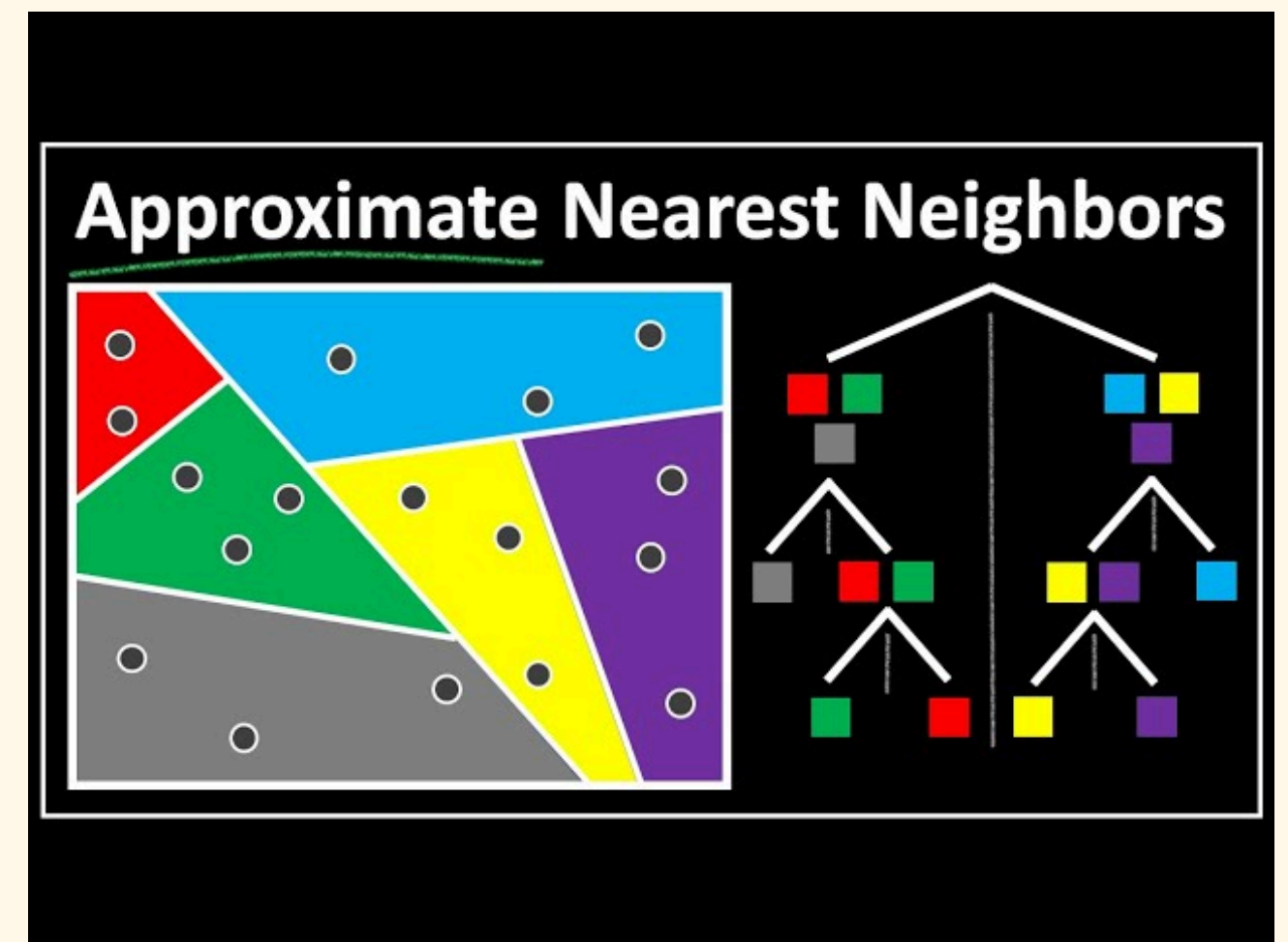
Hexagonal

[2]

Autres : IA & deep learning

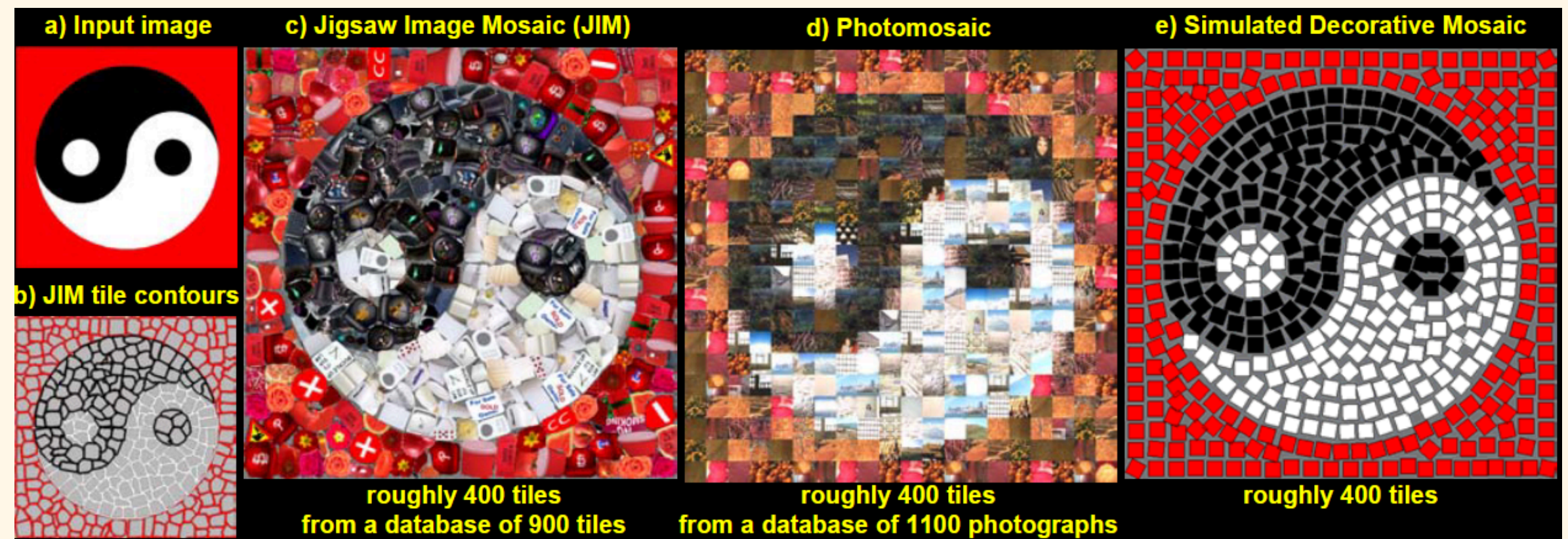
État de l'art

Approximate Nearest Neighbor



[1]

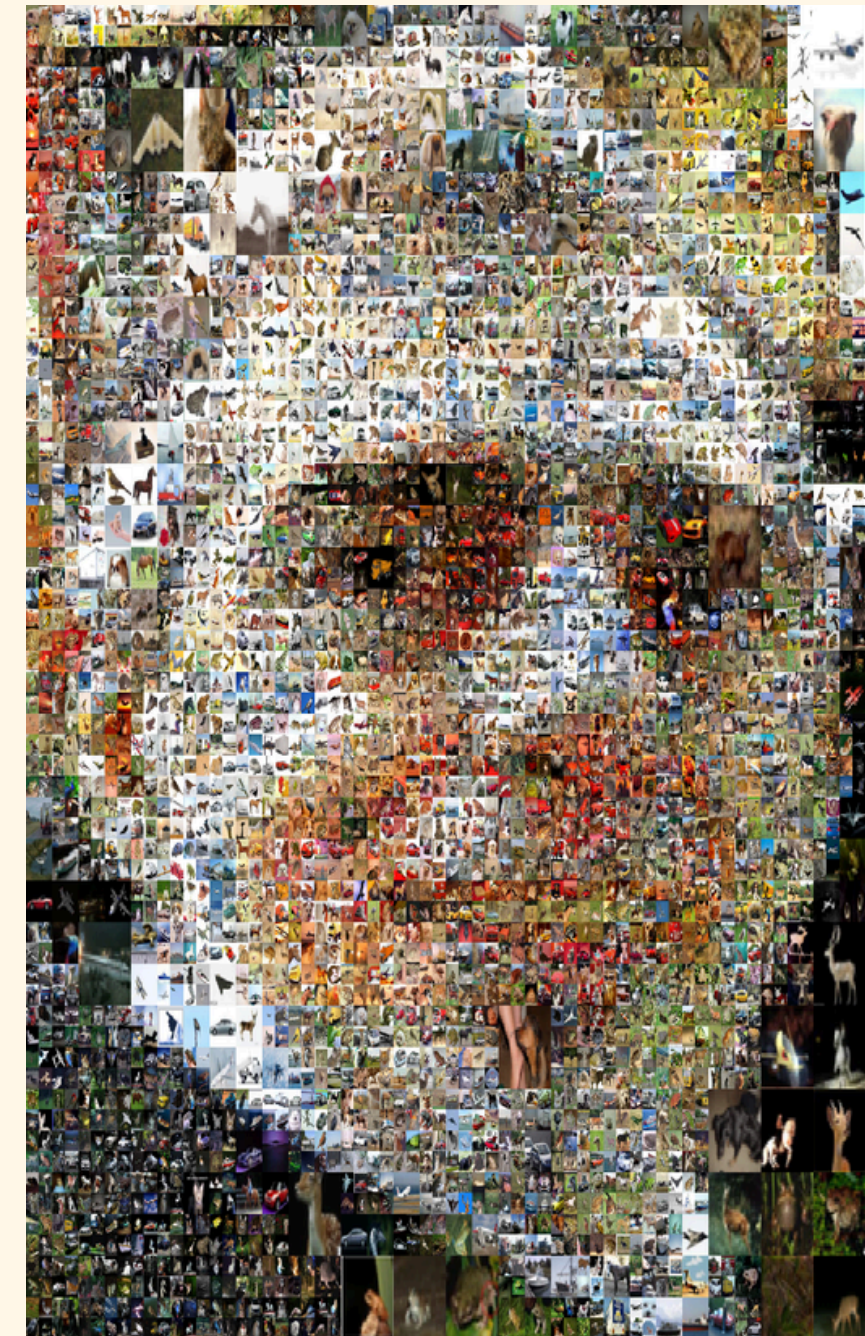
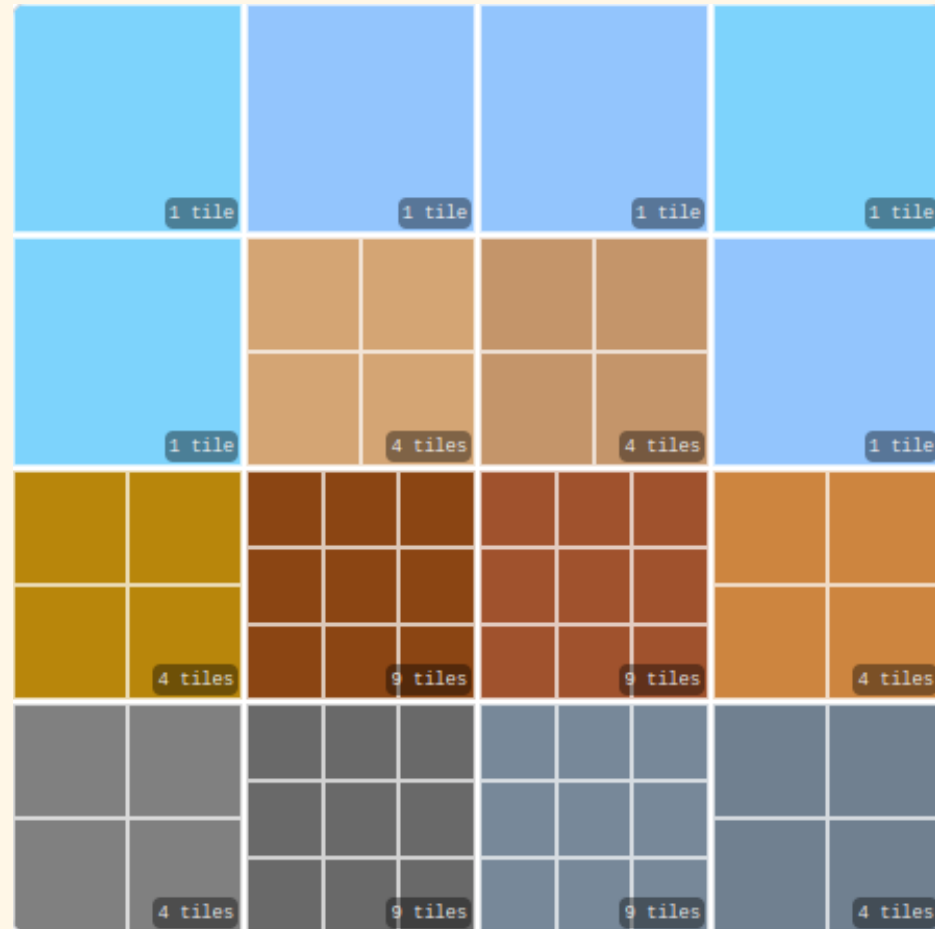
Jigsaw mosaïque



[1]

Méthodes implémentées

Tuiles adaptatives



Seuil de variance : 500

Méthodes implémentées

Super-pixels

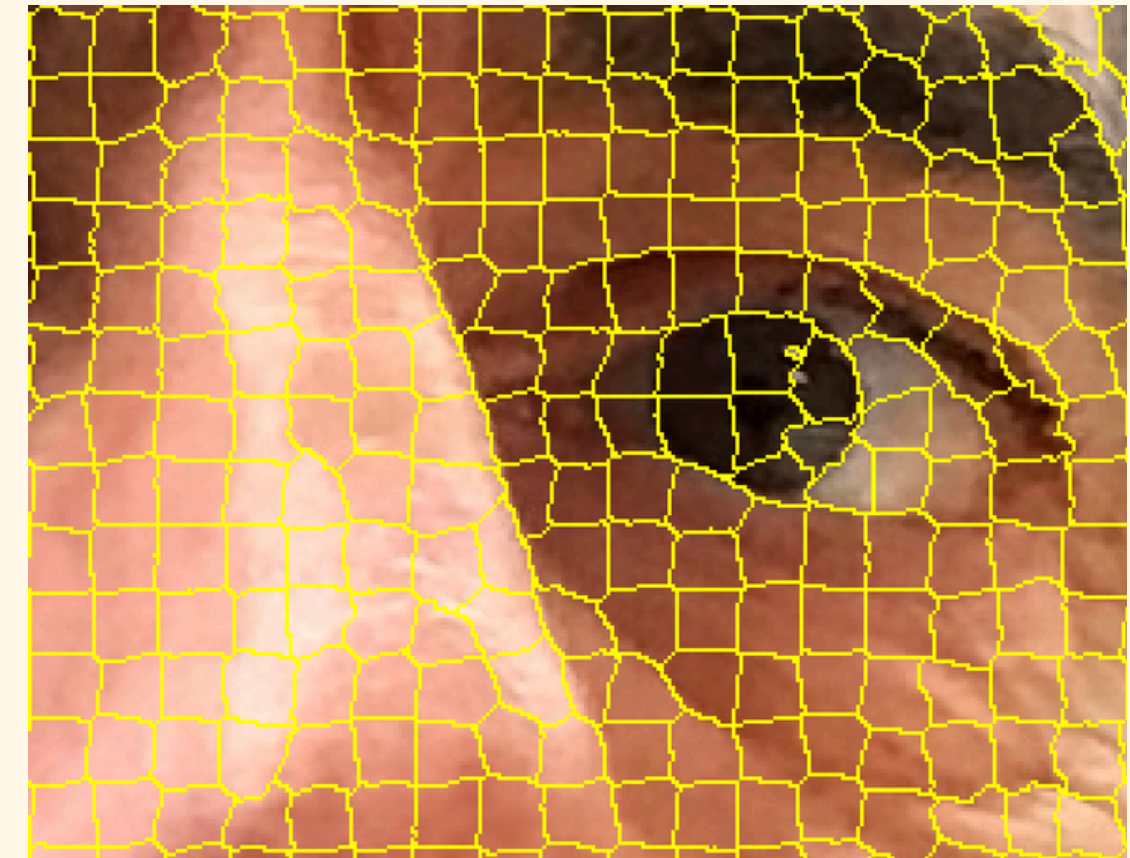


SNIC superpixels



SNIC polygons

[1]



Post-traitement

2nd passe et rotation des imagerie



Première passe



Seconde passe

Résultats

Image en entrée

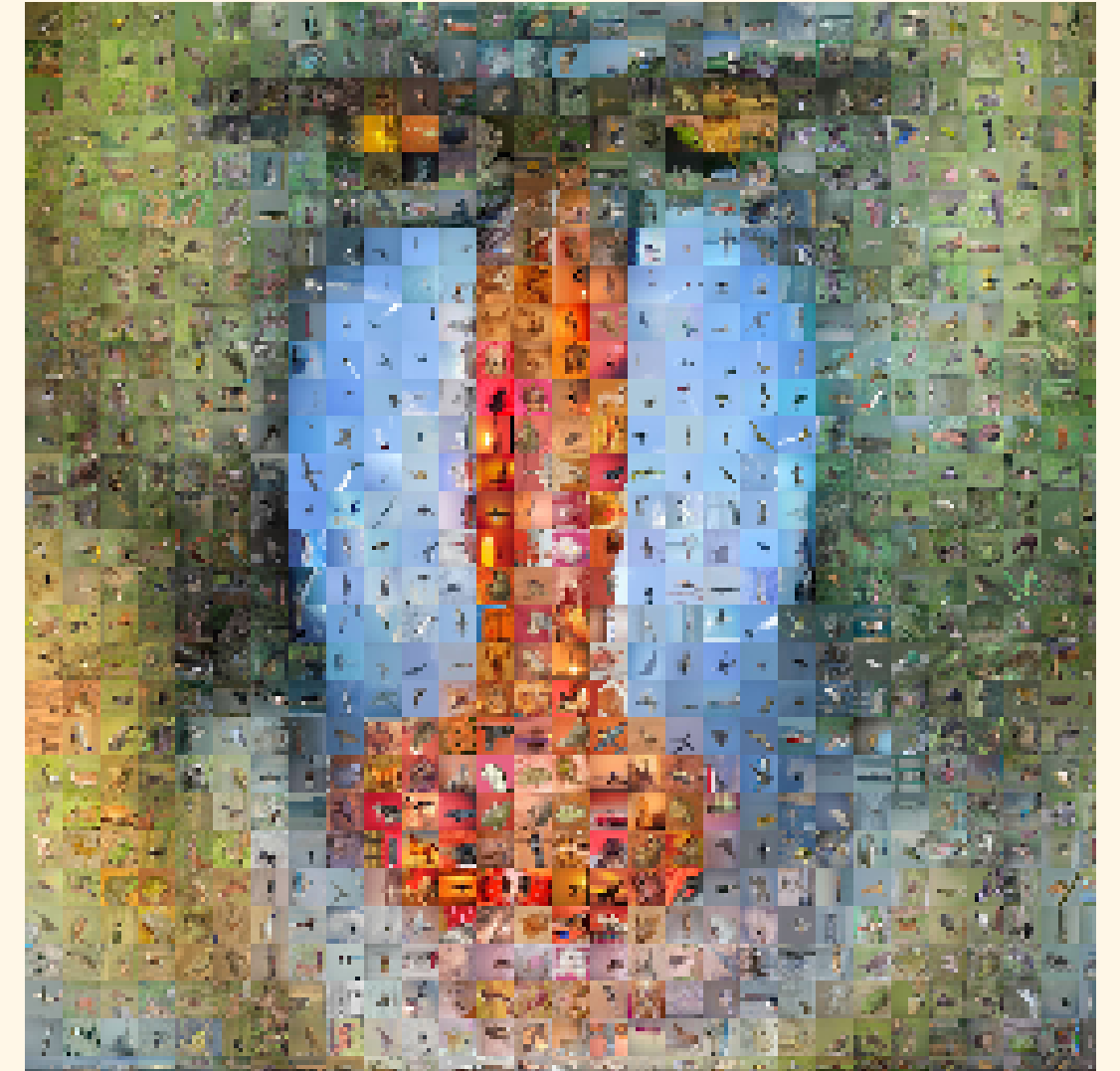


Adaptive tiling

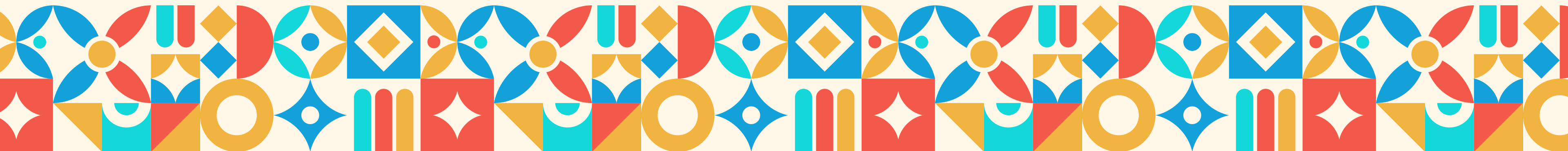


PSNR: 11.77 dB
SSIM: 0.42

Classique

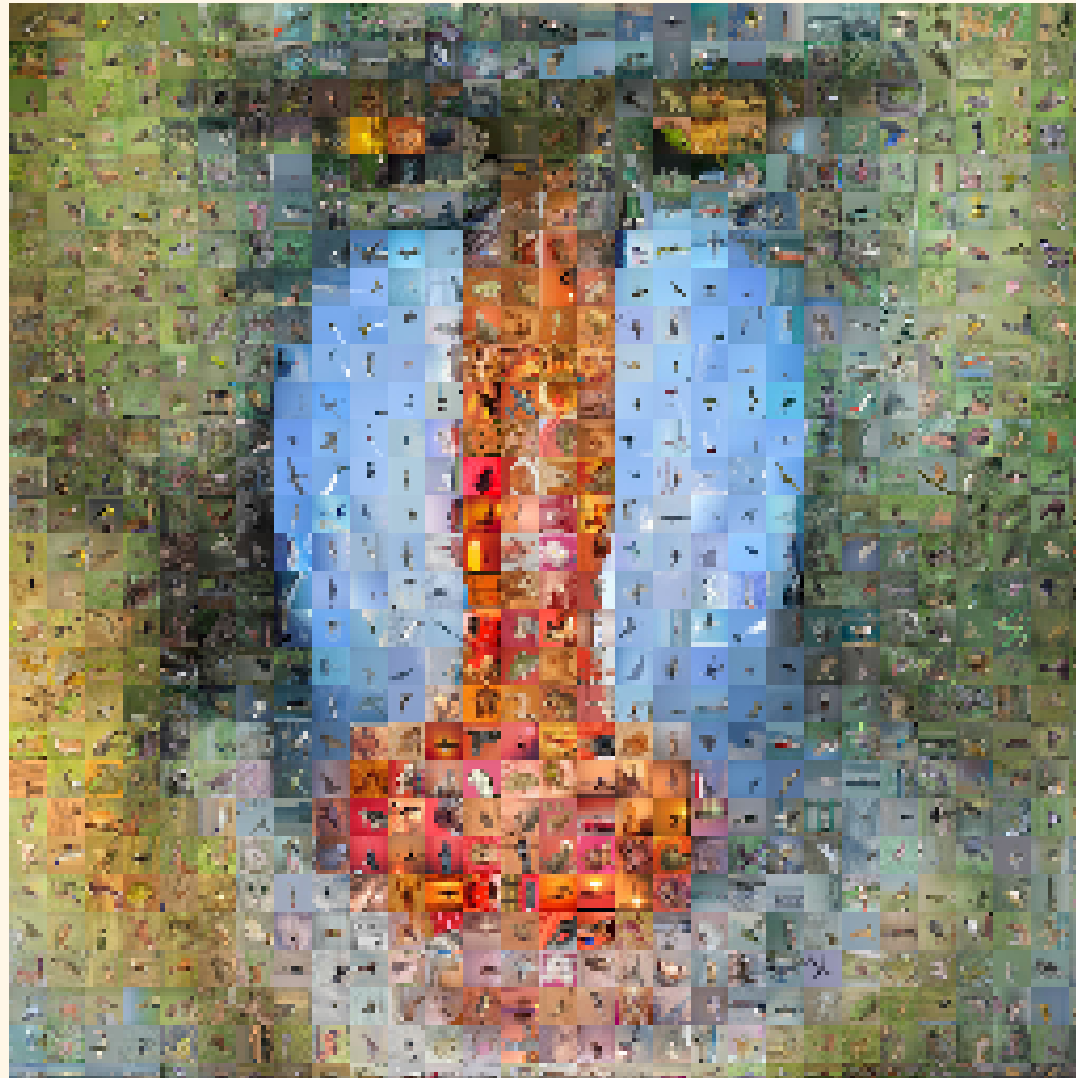


PSNR: 17.56 dB
SSIM: 0.79



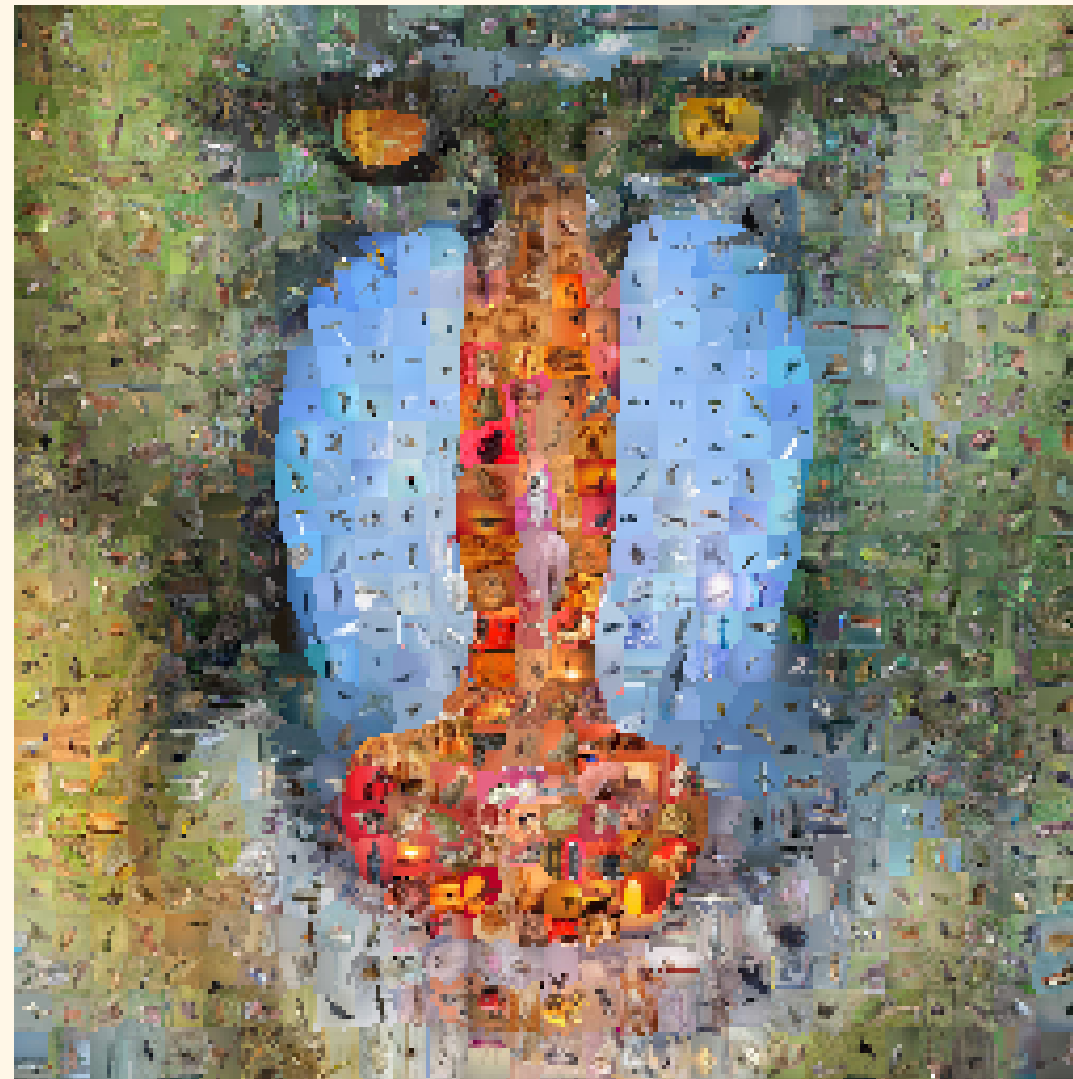
Résultats

Classique + 2nd passe



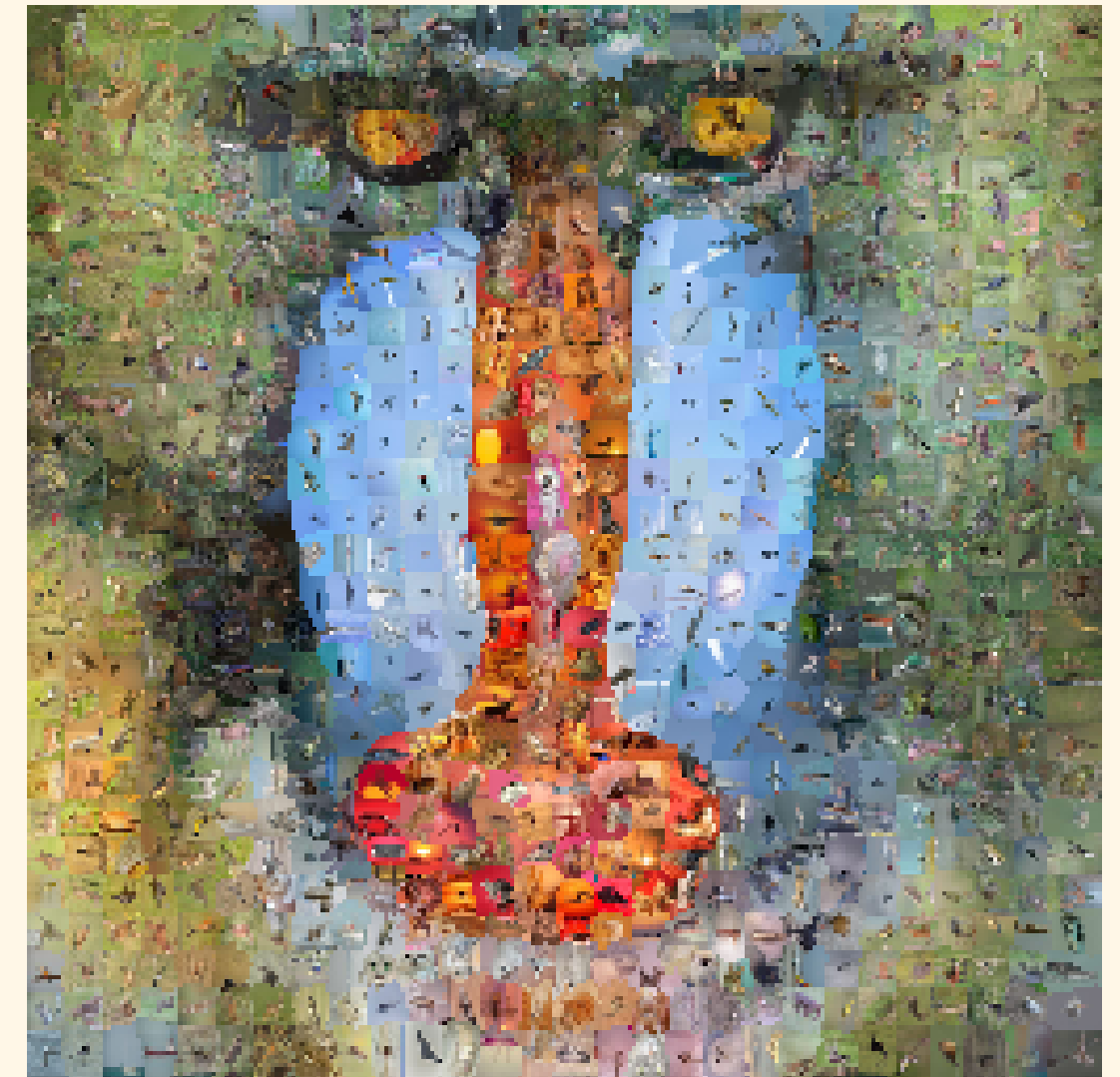
PSNR: 17.83 dB
SSIM: 0.80

Super-pixels

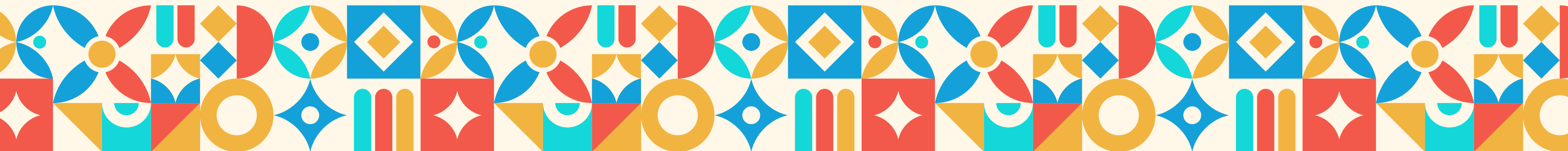


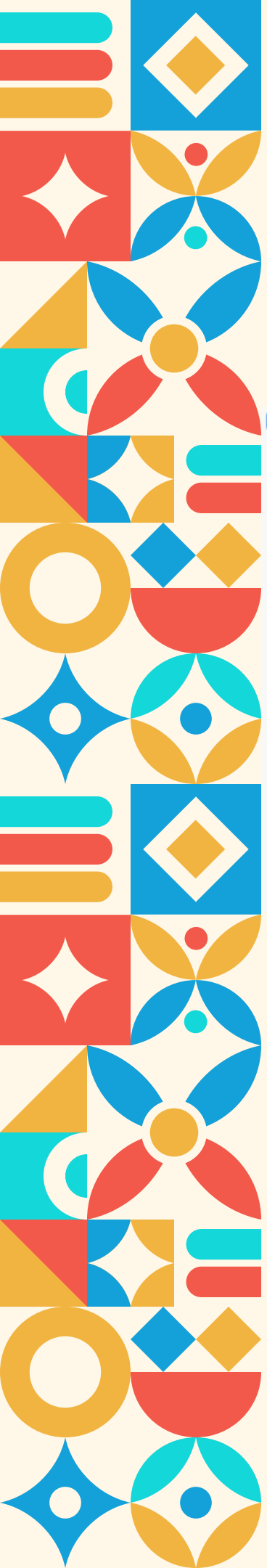
PSNR: 16.65 dB
SSIM: 0.76

Super-pixels + 2nd passe



PSNR: 17.20
SSIM: 0.79





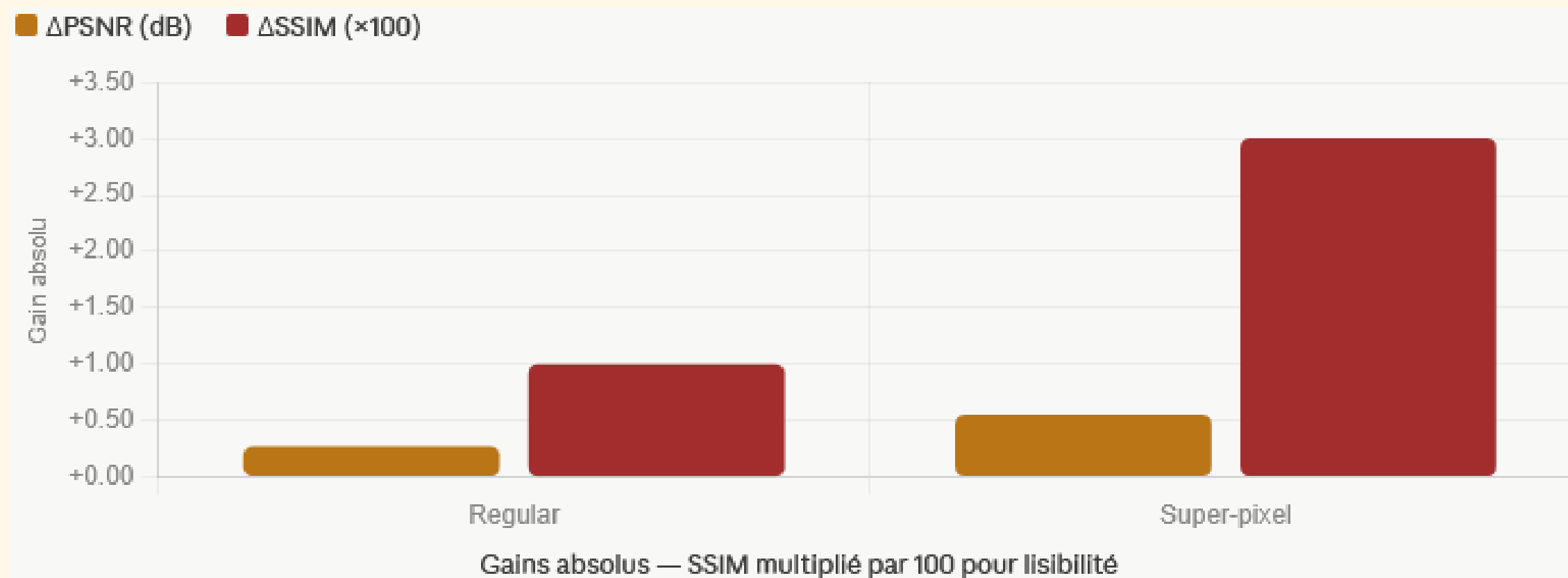
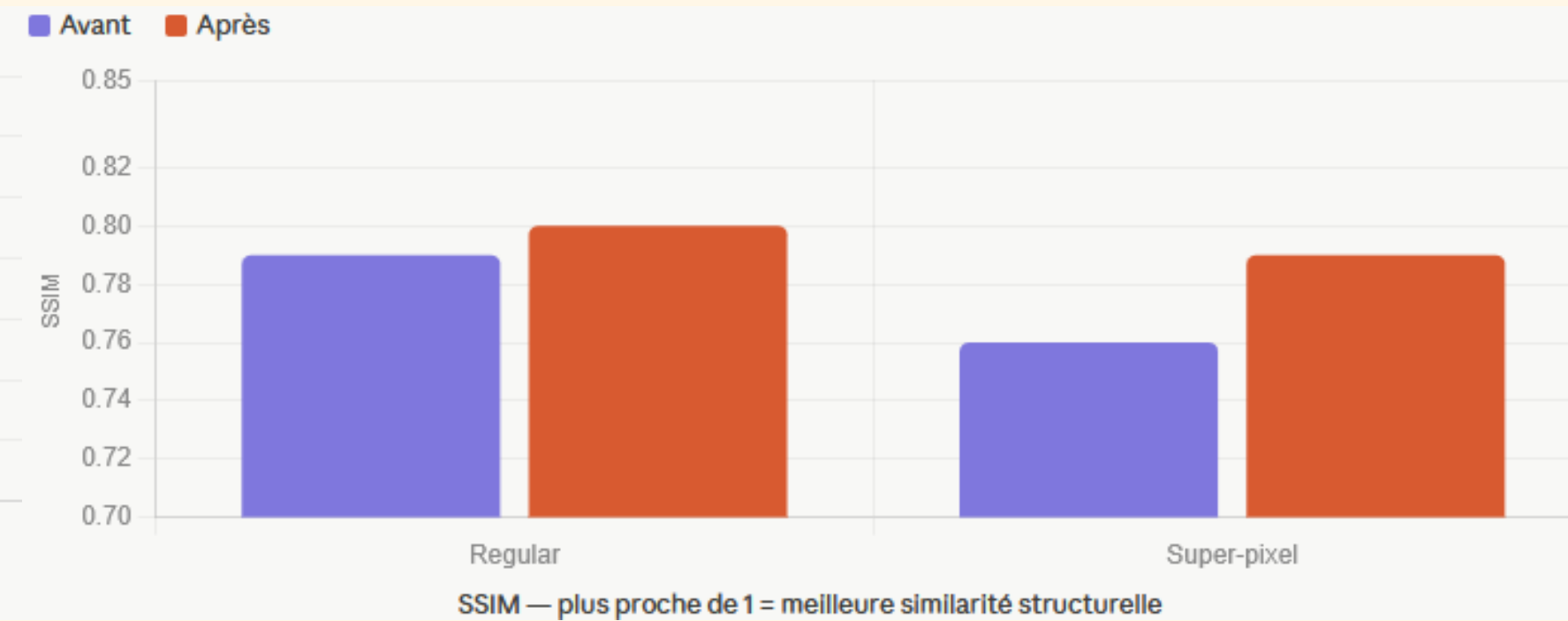
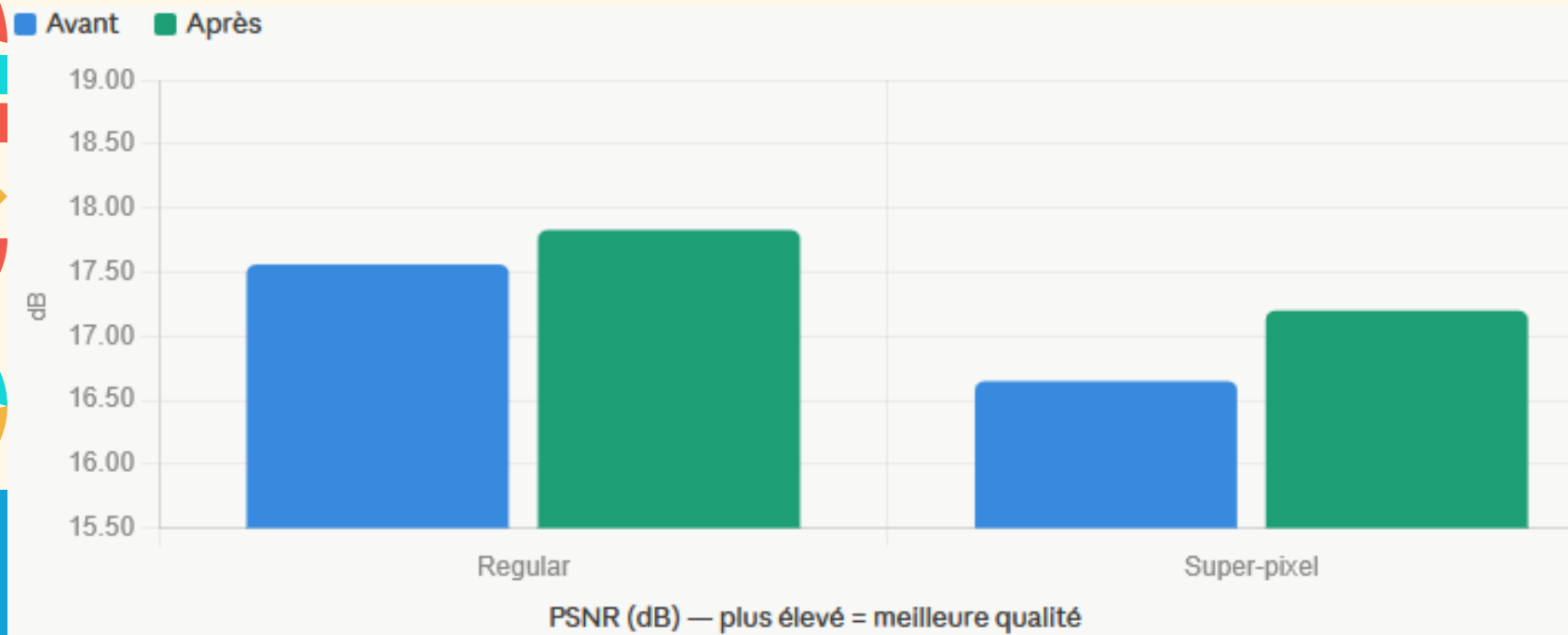
Analyses

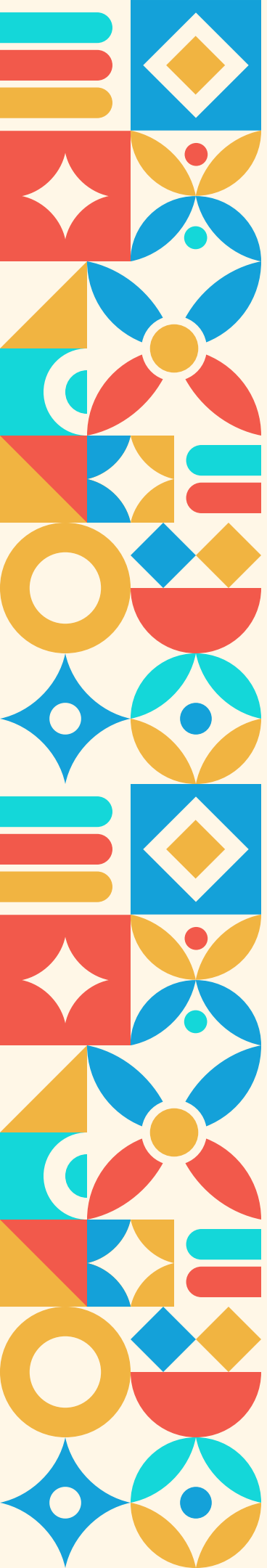
Δ PSNR Regular
+0.27 dB

Δ PSNR Super-pixel
+0.55 dB

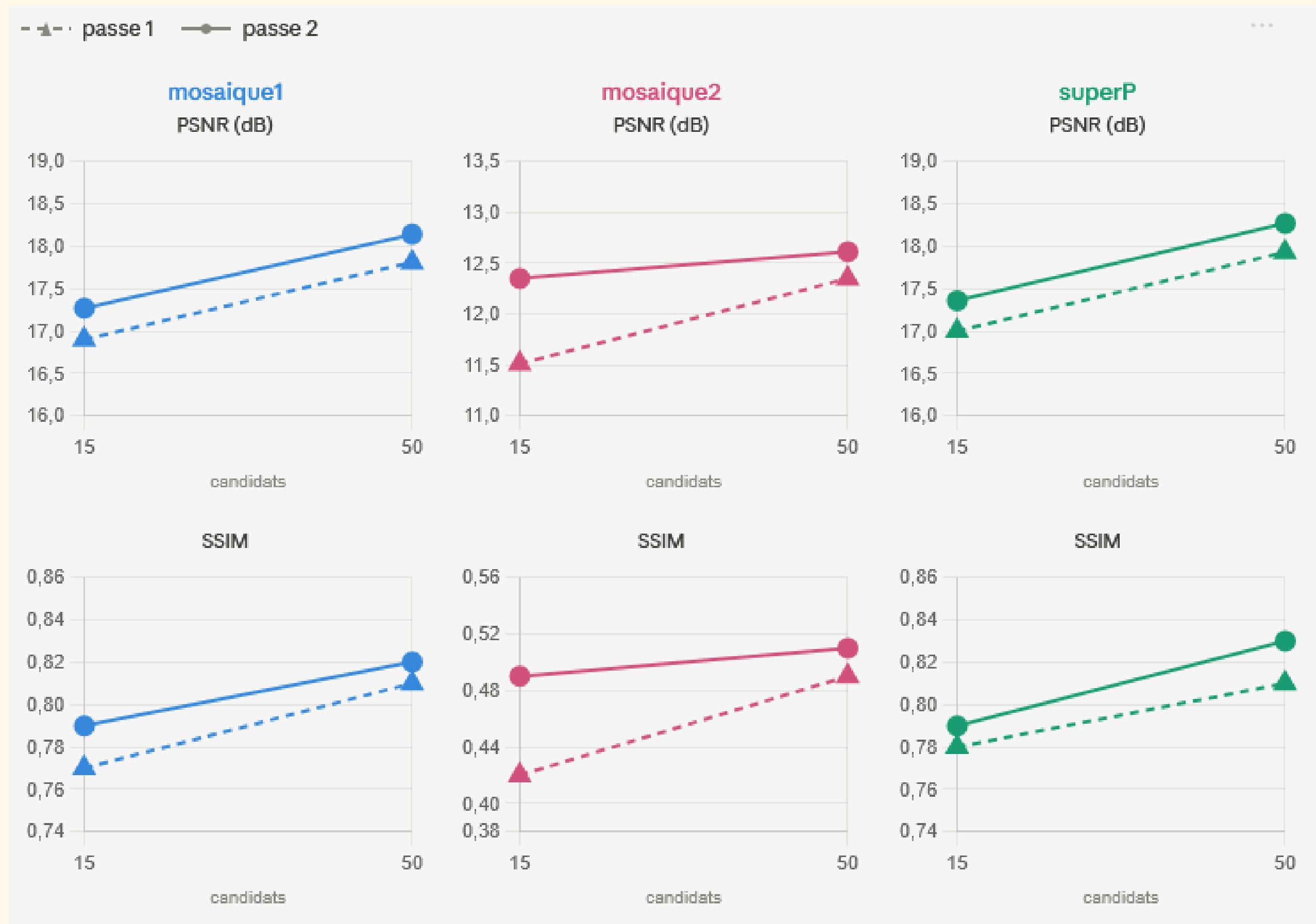
Δ SSIM Regular
+0.010

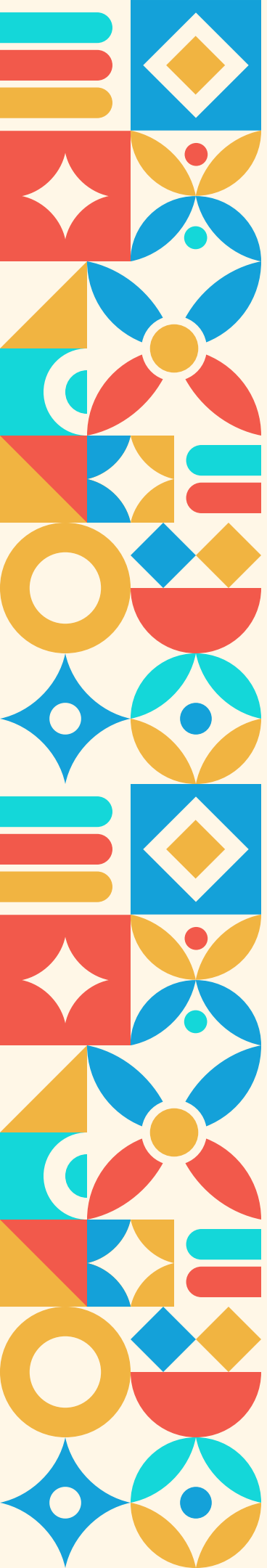
Δ SSIM Super-pixel
+0.030





Analyses





Analyses

Comparaison de résultats entre la mosaïque classique et la super-pixels

**Mosaïque
classique**



**Mosaïque
super-pixels**

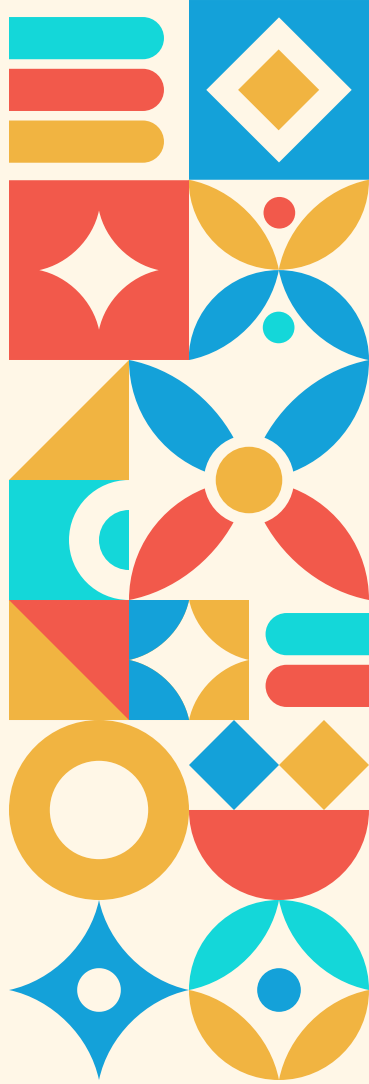
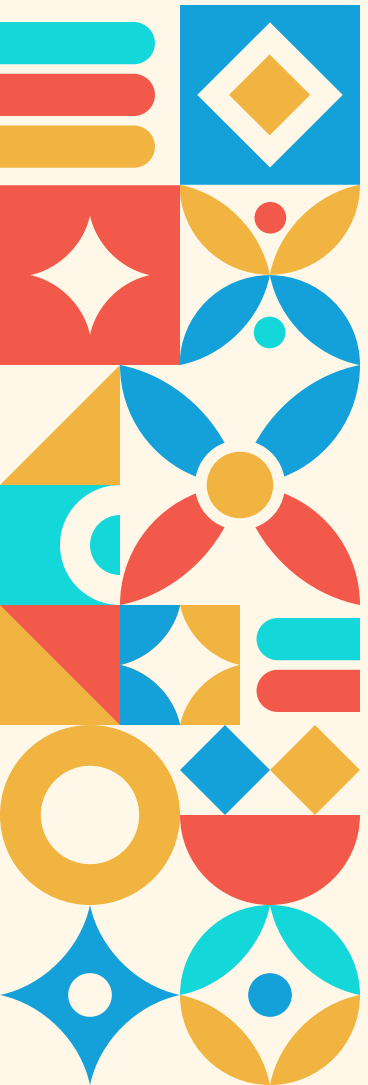


Conclusion

L'adaptive Tiling, bien qu'étant une méthode intéressante, ne produit pas des résultats satisfaisants

La mosaïque classique offre des résultats stables

Le super-pixel est une approche très intéressante dont les résultats métriques sont légèrement inférieurs à la mosaïque standard mais propose un rendu différent



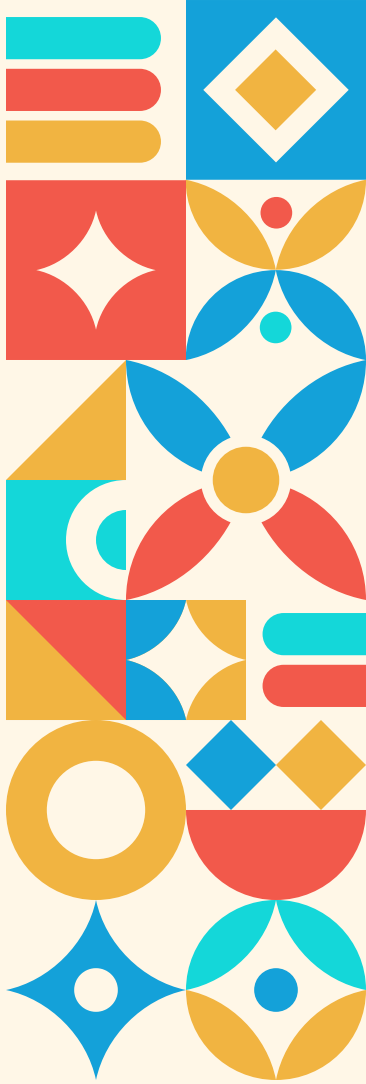
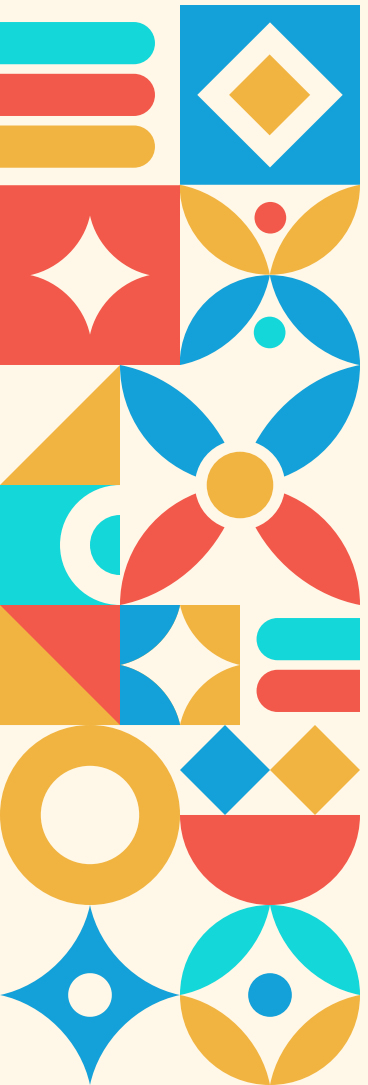
Perspectives

Amélioration de l'adaptive tiling
Optimisation du temps de calcul
Extension des angles de rotation

Interface plus élaborée

Changer d'espace couleurs

Parfaire les comparaisons de la seconde passe



Sources

- [1] <https://www.turbomosaic.com/make-photo-mosaic-in-photoshop.html>
- [2] <https://www.turbomosaic.com/how-to-make-a-photo-mosaic.html>
- [3] <https://www.youtube.com/watch?v=DRbjpuqOsjk>
- [4] J. Kim, F. Pellacini ACM Transactions on Graphics, 2002 - researchgate.net
- [5] R. Achanta, S. Ssstrunk. Superpixels and Polygons using Simple Non-Iterative Clustering. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, pp. 4651-4660
- [6] A. Krizhevsky, V. Nair, G. Hinton. CIFAR-10 dataset, <https://www.cs.toronto.edu/~kriz/cifar.html>

**Merci de
nous avoir
écoutés !**

